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ReRAM resistance design of LRS and HRS for ultrahigh-capacity digital memory and analog Computation-in-Memory

Chihiro MATSUI†, Ayumu YAMADA†, Naoko MISAWA† *Nonmembers and* **Ken TAKEUCHI†**, *Member*

SUMMARY This paper proposes a ReRAM resistance design for ultrahigh-capacity digital memory and analog Computation-in-Memory (CiM). The read-out current of the bit-line is degraded by the interconnection resistance of the bit-line due to IR drop. The bit-line current formulation reveals that the ReRAM resistance should be set as high as $1.0\times10^5\ \Omega$ for the high-capacity digital ReRAM memory. In addition, the ReRAM resistance of LRS and HRS is designed as $1.0\times10^5\ \Omega$ and $1.0\times10^9\ \Omega$ for the high-capacity analog ReRAM CiM.

key words: *Computation-in-Memory, Digital memory, Current-sensing, Interconnection resistance.*

1. Introduction

In the era of data explosion, high-capacity non-volatile memories are being actively developed as a storage [1, 2] and as an accelerator [3-7]. Resistive RAM (ReRAM) is one of the large-capacity non-volatile memories because of its high capacity and low energy [8, 9]. As a storage, ReRAM-based digital memory reads one memory cell connected to a bit-line (BL). On the other hand, analog Computation-in-Memory (CiM) using ReRAM (Fig. 1), which reads multiple memory cells connected to a BL at the same time, is a promising accelerator for multiply-accumulate (MAC) operations of neural network (NN) in machine learning. Input voltage pulses are applied to the memory cells through the source-lines (SLs) while activating the access transistors through the word-lines (WLs). According to Ohm's law and Kirchhoff's current law, the output bit-line (BL) current (I_{BL}) accumulates the cell current from multiple memory cells.

However, the inference accuracy in CiM decreases by the non-idealities of memories such as writing variation of memory cell states, data-retention error, device defects, and short circuits [10-15]. In addition, when the interconnection resistance of BL is present, the BL currents of analog CiM varies and decreases in a complex manner due to IR dop [16-18]. The interconnection resistance of BL severely degrades the read-out BL current, reducing the accuracy of NN [10, 19, 20]. Therefore, the IR drop inhibits scaling of analog CiM using ReRAM. Conventionally, several compensation methods for IR drop are introduced for CiM. The weight matrix of NN, which changes due to the interconnection resistance, is trained or calibrated by considering the interconnection resistance, and the trained or calibrated weight matrix is mapped to memory resistance in CiM. [19, 20]. For memory devices that do not have a large On/Off current ratio, a dummy column is used to remove the offset current by HRS cells from as a MAC result [21]. Contrary to the conventional scaling strategy, reverse scaling is required to increase the interconnection width and reduce the interconnection resistance, thereby reducing IR drop [22]. BL current is approximately formulated based on an intuition that the effect of interconnection resistance becomes smaller as the memory resistance increases. The scaling factor obtained from the approximate BL current reduces the impact of the interconnection resistance [16]. The voltage at each memory cell of CiM matrix is formulated for the case with BL and WL resistances, considering the case with input voltage applied from one or both ends of BL and WL [23].

By focusing only BL resistance of analog CiM, this work gives a simple expression for BL current with input voltage applied from one end. From the BL current formulation, a design for the ReRAM resistance of Low-resistance state (LRS) and high-resistance state (HRS) is proposed for the analog ReRAM CiM. As shown in Fig. 2, an analog TaO_X-based ReRAM cell [24] has a wide range of cell current (I_{CELL}), i.e., a wide range of resistance by changing write current. LRS and HRS carry large and small I_{CELL},

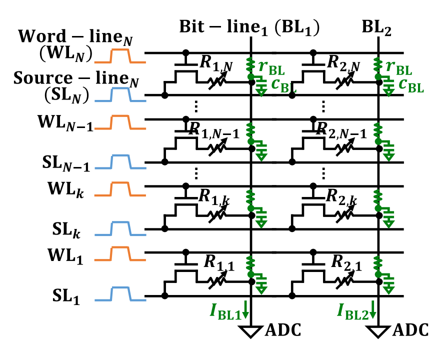


Fig. 1 1T1R analog Computation-in-Memory (CiM) with ReRAM cells.

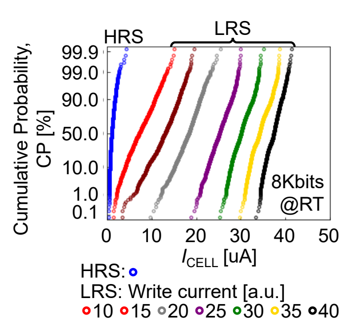


Fig. 2 Measured ReRAM cell current (I_{CELL}) [25].

† The University of Tokyo, Tokyo, 113-8656 Japan.

respectively. The resistances of LRS are set from HRS by applying write current. The ReRAM resistances of LRS and HRS by appropriate write currents are stored in ReRAM CiM cells. The reminder of this paper is organized as follows.

- Sect. 2 formulates BL current of ReRAM-based digital memory and analog CiM. The formulation finds that the BL current of analog ReRAM CiM varies more complexly than that of digital ReRAM memory, due to the BL resistance.
- In Sect. 3, the resistance of LRS, which flows larger cell current than HRS, is designed for both ReRAM-based digital memory and analog CiM.
- Sect. 4 considers design step of LRS and HRS resistance for analog ReRAM CiM, which are less sensitive to IR drop of BL and provides the required signal BL current.
- Finally, Sect. 5 summarizes the resistance design of ReRAM-based digital memory and analog CiM.

2. Bit-line Current Formulation for ReRAM-based Digital Memory and Analog Computation-in-Memory

Circuit models of (a) digital memory and (b) analog CiM using ReRAM shown in Fig. 3 have N ReRAM memory cells connected between a pair of SL and BL. For simplicity, different from the actual circuit in Fig. 1, the access transistors are removed from the circuit models. All N ReRAM memory resistances and the interconnection resistances of BL between memory cells are assumed to be constant at R and r_{BL}, respectively. Even in the future scaled digital memory and analog CiM using ReRAM, r_{BL} is assumed to be smaller than R. The SL voltage applied to the ReRAM cells is also assumed to be constant at V_{SL}. To discuss parallel operation along the BL direction, the following formulation ignores the resistance of SL which is orthogonal to the BL. HSPICE simulation is also performed to verify equations of digital memory and analog CiM using ReRAM.

2.1 Bit-line Current of Digital ReRAM Memory

In the digital ReRAM memory circuit (Fig. 3(a)), only the k-th memory cell from the end of the BL is read out at a time by activating only k-th SL (SL_k) to V_{SL}. In this situation, the cell current (i_k) flows through the k-th memory cell (R_k), while current does not flow through the other memory cells (R_N, $R_{N\text{-}1}$, …, R_{k+1}, $R_{k\text{-}1}$, …, R_2, and R_1). The local BL current j_k equals to the local BL current on the ADC side ($j_{k\text{-}1}$, …, j_2, and j_1). The local BL currents on the opposite side of the ADC (j_N, $j_{N\text{-}1}$, …, and j_{k+1}) do not flow. In the ideal situation of $r_{\mathrm{BL}} = 0\ \Omega$, the BL current of the digital ReRAM memory is shown in Eq. (1).

$$I_{\mathrm{BL\_mem\_ideal}} = \frac{V_{\mathrm{SL}}}{R} \tag{1}$$

As the actual situation of presenting the BL resistance, the equivalent resistance becomes the series connection of

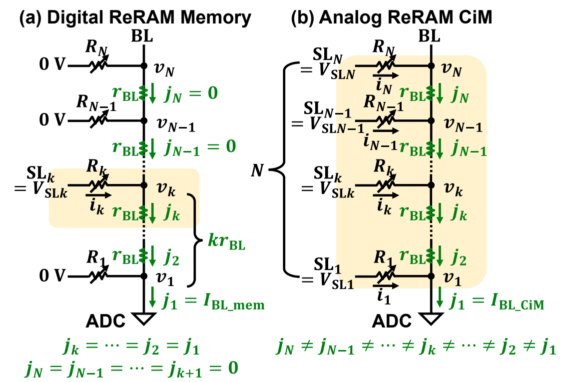


Fig. 3 Circuit models for BL current formulation. (a) Digital ReRAM memory which reads only k-th memory cell. (b) Analog ReRAM CiM which reads multiple N-memory cells connected to BL.

memory resistance R and k BL resistance r_{BL}. As in Eq. (2), k consecutive BL resistances ($k \times r_{\mathrm{BL}}$) degrade the BL current $I_{\mathrm{BL\_mem}}$ of actual situation.

$$I_{\mathrm{BL\_mem}} = \frac{V_{\mathrm{SL}}}{R + kr_{\mathrm{BL}}} \tag{2}$$

Fig 4 shows $I_{\mathrm{BL\_mem}}$ analyzed using Eq. (2) when reading the k-th memory cell in the digital ReRAM memory. The large r_{BL} and small memory resistance R decrease $I_{\mathrm{BL\_mem}}$ from the ideal value, as shown in Figs. 4(a) and 4(b). The ideal BL current is defined as that with $r_{\mathrm{BL}} = 0\ \Omega$. When $r_{\mathrm{BL}} = 0\ \Omega$, $I_{\mathrm{BL\_mem\_ideal}}$ becomes a straight line. Here, V_{SL} is set to 0.4 V. $I_{\mathrm{BL\_mem}}$ with $r_{\mathrm{BL}} = 0.1\ \Omega$ (1 Ω) decreases by 9.3% (50.6%) from $I_{\mathrm{BL\_mem\_ideal}}$ when reading $k = 1024$th memory cell of $R = 10^3\ \Omega$. If reading $R = 10^9\ \Omega$ at $k = 1024$th memory cell, $I_{\mathrm{BL\_mem}}$ with $r_{\mathrm{BL}} = 0.1\ \Omega$ (1 Ω) decreases by 0.00001% (0.0001%). The high memory cell resistance R does not reduce the BL current $I_{\mathrm{BL\_mem}}$ even in the presence of the BL resistance r_{BL}. The length (L), width (W), and transconductance (g_{m}) of the access transistor used in the following HSPICE simulations as verification are 0.18 um, 2 um, and 0.92 mS, respectively. In HSPICE simulations (Fig. 4(c)), $I_{\mathrm{BL\_mem}}$ is degraded by small R when $r_{\mathrm{BL}} = 1\ \Omega$. $I_{\mathrm{BL\_mem}}$ by HSPICE simulation degrades more than that by Eq. (2) due to the access transistor of 1T1R ReRAM cell.

2.2 Bit-line Current of Analog ReRAM CiM

The analog ReRAM CiM in Fig. 3(b) activates multiple SLs to V_{SL} so that the cell current flows simultaneously to the multiple ReRAM memory cells (R_1, …, R_k, …, $R_{N\text{-}1}$, R_N) and reads out the BL current. Different from the digital ReRAM memory, in the analog ReRAM CiM, the cell current (i_N, $i_{N\text{-}1}$, …, i_k, …, and i_1) flows through each memory cell, and the local BL current (j_N, $j_{N\text{-}1}$, …, j_k, …, and j_1) accumulates cell currents. The BL current $I_{\mathrm{BL\_CiM}}$ of the analog ReRAM CiM is detected as j_1 at the end of BL, where the ADC is connected. When $r_{\mathrm{BL}} = 0\ \Omega$ as an ideal case, the BL current $I_{\mathrm{BL\_CiM\_ideal}}$ of the analog ReRAM CiM is obtained with the N-parallel memory cells (Eq. (3)).

$$I_{\mathrm{BL\_CiM\_ideal}} = \frac{NV_{\mathrm{SL}}}{R} \tag{3}$$

However, the local BL voltage v_k is not constant along BL if the BL resistance r_{BL} is present as an actual situation. The local BL voltage v_k drops by r_{BL}. The next three equations (4-6) are formulated with the assumption that the cell current i_k flows through the k-th memory cell, and the local BL current j_k accumulates the cell currents.

$$j_k - j_{k+1} = i_k \tag{4}$$

$$v_{k+1} - v_k = r_{\mathrm{BL}} j_{k+1} \tag{5}$$

$$V_{\mathrm{SL}} - v_k = R i_k \tag{6}$$

From Eqs. (4-6), the total BL current $I_{\mathrm{BL\_CiM}}$ ($= j_1$) is obtained as in Eq. (7) with the parameters in Eq. (8).

$$I_{\mathrm{BL\_CiM}} = j_1 = -\frac{\lambda_1^N - \lambda_2^N}{(\lambda_2 - 1)\lambda_1^N - (\lambda_1 - 1)\lambda_2^N} \frac{V_{\mathrm{SL}}}{R} \tag{7}$$

$$\lambda_1 = \frac{(\gamma - \delta + 2)}{2}, \quad \lambda_2 = \frac{(\gamma + \delta + 2)}{2}, \quad \delta = \sqrt{\gamma^2 + 4\gamma}, \quad \gamma = \frac{r_{\mathrm{BL}}}{R} \tag{8}$$

Here, $\gamma = r_{\mathrm{BL}}/R$ indicates the resistance ratio of the BL resistance and the memory resistance. The derivation of Eqs. (7) and (8) is shown in Appendix.

Fig. 5 shows $I_{\mathrm{BL\_CiM}}$ analyzed using Eq. (7) and in HSPICE simulations when activating N multiple memory cells at once in the analog ReRAM CiM. When the number of activating memory cells N is large, $I_{\mathrm{BL\_CiM}}$ becomes large by accumulating cell currents from N cells. $I_{\mathrm{BL\_CiM\_ideal}}$ becomes a straight line when $r_{\mathrm{BL}} = 0\ \Omega$. When r_{BL} presents, $I_{\mathrm{BL\_CiM}}$ decreases significantly as N increases. In addition, $I_{\mathrm{BL\_CiM}}$ is reduced at large $\gamma = r_{\mathrm{BL}}/R$, i.e., large r_{BL} and small R. From Figs. 5(a) and 5(b), $I_{\mathrm{BL\_CiM}}$ with $r_{\mathrm{BL}} = 0.1\ \Omega$ (1 Ω) decreases by 90.2% (96.9%) from $I_{\mathrm{BL\_CiM\_ideal}}$ when activating 1024 memory cells of $R = 10^3\ \Omega$. If reading $R = 10^9\ \Omega$ of $N = 1024$ memory cells at once, $I_{\mathrm{BL\_CiM}}$ with $r_{\mathrm{BL}} = 0.1\ \Omega$ (1 Ω) decreases by 0.0035% (0.035%). Same as the digital ReRAM memory, the high memory cell resistance R does not reduce the BL current $I_{\mathrm{BL\_CiM}}$ even in the presence of the BL resistance r_{BL}. In HSPICE simulation results (Fig. 5(c)), $I_{\mathrm{BL\_CiM}}$ is decreased more from $I_{\mathrm{BL\_CiM\_ideal}}$ due to the access transistor of 1T1R ReRAM cell.

The expression for $I_{\mathrm{BL\_CiM}}$ in Eq. (7) is approximated for $N >> 1$. Although λ_1 is always less than 1, the parameter λ_1 becomes much less than 1 when the ratio γ of BL resistance r_{BL} and memory resistance R becomes close to 1, which means the value of r_{BL} is close to that of R. In contrast, λ_2 is always larger than 1. Thus, the number N of parallel memory cells to one BL is large, λ_1^N becomes almost 0. In short, if $N >> 1$, then $\lambda_1^N \approx 0$ since $\lambda_1 < 1$. Therefore, in Eq. (7), the term containing λ_1^N is ignored, and the remaining λ_2^N is cancelled. In this case, a simple approximate equation for $I_{\mathrm{BL\_CiM}}$, which are independent of the number of memory cells N, are obtained in Eq. (9).

$$I_{\mathrm{BL\_CiM}} = j_1 \approx \frac{1}{1 - \lambda_1} \frac{V_{\mathrm{SL}}}{R} \tag{9}$$

Eq. (9) indicates that when N is large, the number of BL resistance r_{BL} degrades the BL current of analog ReRAM CiM. As a result, $I_{\mathrm{BL\_CiM}}$ saturates by the effect of the large number of r_{BL} and does not detect the number of inputs N.

3. ReRAM LRS Resistance Design of Digital ReRAM Memory and Analog ReRAM CiM

The design of the ReRAM LRS resistance R_{LRS} is considered by using Eqs. (1-3) and (7) of the BL current. As shown in Figs. 4 and 5, the small memory cell R carries large BL current, but the presence of BL resistance r_{BL} decreases the BL current accumulated through the small memory cell R. With an indicator α of the proximity, Eq. (10) defines the actual BL current I_{BL} and the ideal BL current $I_{\mathrm{BL\_ideal}}$ for the digital ReRAM memory and the analog ReRAM CiM.

$$I_{\mathrm{BL}} = \alpha I_{\mathrm{BL\_ideal}}, \quad 0 < \alpha < 1 \tag{10}$$

The indicator α is set to about 0.99 because a previous work [26] reports that the MNIST inference accuracy is maintained even with 1.6% error on a Multi-layer

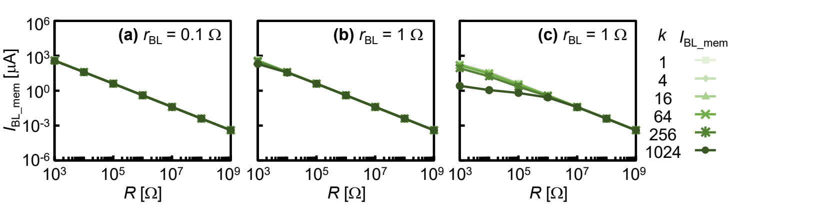


Fig. 4 BL current ($I_{\mathrm{BL\_mem}}$) analysis of digital ReRAM memory with (a) r_{BL} =0.1 Ω and (b) r_{BL} = 1 Ω by Eq. (2). (c) $I_{\mathrm{BL\_mem}}$ with r_{BL} = 1 Ω by HSPICE simulations.

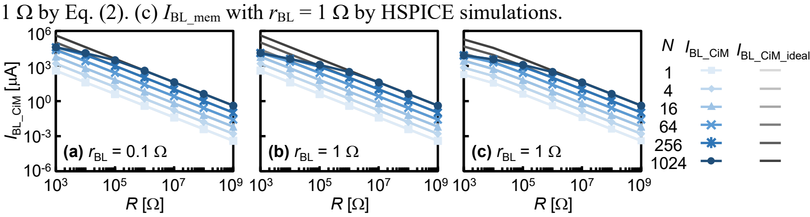


Fig. 5 BL current ($I_{\mathrm{BL\_CiM}}$) analysis of analog ReRAM CiM with (a) r_{BL} =0.1 Ω and (b) r_{BL} = 1 Ω by Eq. (7). (c) $I_{\mathrm{BL\_CiM}}$ with r_{BL} = 1 Ω by HSPICE simulations.

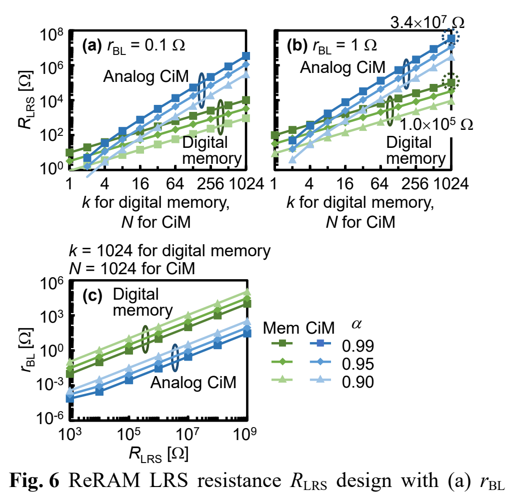


Fig. 6 ReRAM LRS resistance R_{LRS} design with (a) r_{BL} =0.1 Ω, (b) r_{BL} =1 Ω, and (c) BL resistance r_{BL} design for ReRAM-based digital memory and analog CiM obtained by Eq. (10).

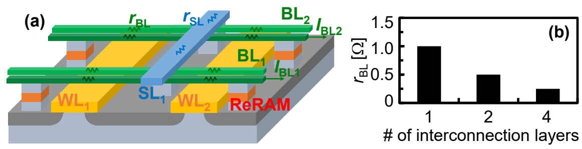


Fig. 7 (a) 1T1R ReRAM cell and (b) BL resistance (r_{BL}) reduction by multi-level interconnection with 22 nm Cu interconnects.

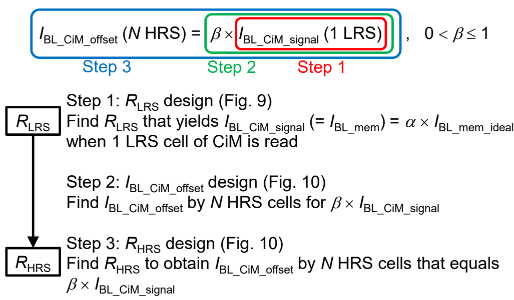


Fig. 8 Design steps of ReRAM resistance for analog ReRAM CiM.

Perceptron.

Figs. 6(a) and 6(b) compare the ReRAM LRS resistance R_{LRS} to obtain αI_{BL_ideal} between the digital ReRAM memory and the analog ReRAM CiM with r_{BL} set to 0.1 Ω and 1 Ω. When the memory number k to be read is 1024 in the digital ReRAM memory, the ReRAM LRS resistance R_{LRS} should be set as high as 1.0×10^5 Ω with r_{BL} = 1 Ω and α = 0.99. When the number of connected memory cells N is 1024 in the analog ReRAM CiM, R_{LRS} should be as high as 3.4×10^7 Ω with r_{BL} = 1 Ω and α = 0.99. When k and N are the same number, the memory resistance R_{LRS} of the analog ReRAM CiM must be higher than that of the digital ReRAM memory. Because the cell current i_k decreases with large k, the memory resistance R_{LRS} should be high to keep $\gamma = r_{BL}/R_{LRS}$ small. When the indicator α is smaller, the memory resistance R_{LRS} can be smaller for both the digital ReRAM memory and the analog ReRAM CiM.

In addition, Fig. 6(c) shows the acceptable BL resistance r_{BL} obtained by Eq. (10). The higher the memory resistance R_{LRS}, the higher the BL resistance r_{BL} is tolerated for both the digital ReRAM memory and the analog ReRAM CiM. From Figs. 6(a) and 6(b), when $\alpha = 0.99$, the digital ReRAM memory and the analog ReRAM CiM require a resistance ratio of $\gamma = r_{BL}/R_{LRS} \approx 10^{-5}$ and 3×10^{-8}, respectively. For the same memory resistance R_{LRS}, the analog ReRAM CiM requires a lower BL resistance r_{BL} than the digital ReRAM memory (Fig. 6(c)). The lower BL resistance r_{BL} is achieved by high number of multi-level interconnections (Fig. 7). If the two or more BLs are connected by TSV in places, the BL resistance r_{BL} is reduced by making the BL parallel. If r_{BL} = 1 Ω of 22 nm Cu single-level interconnects, r_{BL} would be 0.5 Ω and 0.25 Ω with double- and quadrable-level interconnections, respectively.

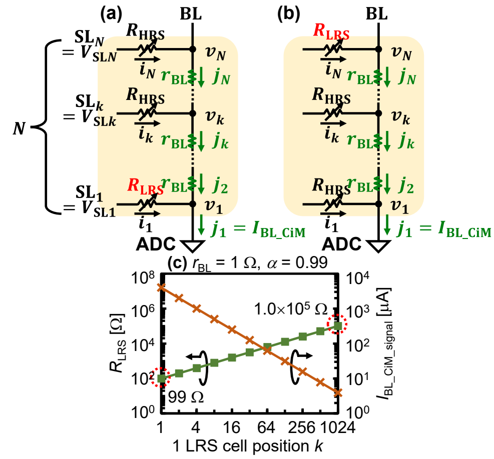


Fig. 9 (Step 1 R_{LRS} design) Resistance pattern in analog ReRAM CiM with N memory cells. 1 LRS cell stored at (a) k = 1 (best) and (b) $k = N$ (worst) cases. Other cells store HRS. (c) R_{LRS} and $I_{BL\_CiM\_signal}$ by 1 LRS cell at different LRS cell position in analog ReRAM CiM.

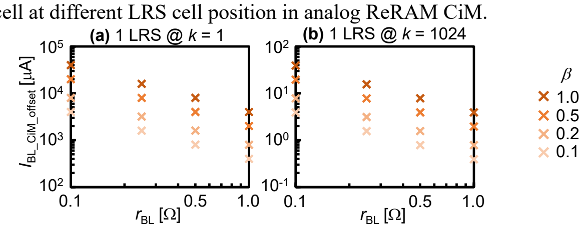


Fig. 10 (Step 2: $I_{BL\_CiM\_offset}$ design) $I_{BL\_CiM\_offset}$ with β by 1024 HRS cells when 1 LRS positioned at (a) k = 1 (best) and (b) k = 1024 (worst) cases.

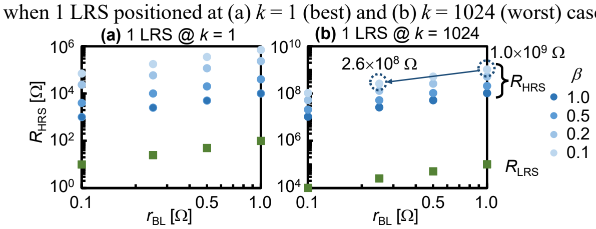


Fig. 11 (Step 3: R_{HRS} design) R_{HRS} design when 1 LRS positioned at (a) k = 1 (best) and (b) k = 1024 (worst) cases. R_{HRS} values should be higher than points in graph to eliminate $I_{BL\_CiM\_offset}$.

4. ReRAM LRS and HRS Resistance Design of Analog ReRAM CiM

The ReRAM memory cells in the analog ReRAM CiM store LRS as "1" and HRS as "0" of the weights in the neural network. The offset current by HRS cells should be much lower than the signal current by LRS cells in the analog ReRAM CiM [21]. As a practical case, the offset current $I_{BL\_CiM\_offset}$ by N HRS cells should be β-times smaller than the signal current $I_{BL\_CiM\_signal}$ by 1 LRS cell as in Eq. (11).

$$I_{BL\_CiM\_offset} = \beta I_{BL\_CiM\_signal}, \quad 0 < \beta \leq 1 \qquad (11)$$

The ReRAM resistance of LRS and HRS cells in the analog ReRAM CiM is designed by three steps as shown in Fig. 8. In Step 1, given r_{BL}, the LRS resistance R_{LRS} is found that yields α of the BL current value ($I_{BL\_CiM\_signal}$) as a signal when reading 1 LRS cell in one BL. I_{BL_mem} value is used as $I_{BL\_CiM\_signal}$ because one cell reading in the analog ReRAM CiM is equivalent to one cell reading in the digital ReRAM memory. In Step 2, $I_{BL\_CiM\_offset}$ is found when N HRS cells are parallel read in the analog ReRAM CiM. The cell current from HRS cells, which is not unwanted, is considered as an offset of the BL current. Then, in Step 3, the HRS resistance R_{HRS} is obtained. R_{HRS} yields $I_{\mathrm{BL\_CiM\_offset}}$ that is β times smaller than $I_{\mathrm{BL\_CiM\_signal}}$. As a result, the signal current $I_{\mathrm{BL\_CiM\_signal}}$ by LRS cell can be distinguished from the offset current $I_{\mathrm{BL\_CiM\_offset}}$ by HRS cells. Figs. 9-11 shows the details of the design steps for the number of memory cells N up to 1024.

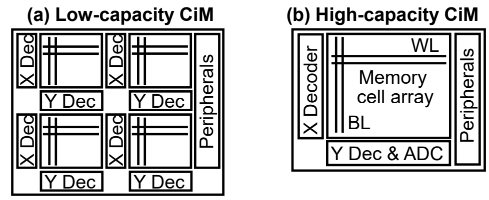
Step 1: R_{LRS} design (Fig. 9) When N = 1024, the 1 LRS ReRAM cell is stored at k = 1 in the best case and at k = 1024 in the worst case (Figs. 9(a) and 9(b)). The memory cell furthest from the ADC (k = 1024) is most affected by the BL resistance r_{BL}. Fig. 9(c) shows the relationship of RLRS and $I_{\mathrm{BL\_CiM\_signal}}$ when reading different memory cell position k in the analog ReRAM CiM. Same as Fig. 6(b), from Eqs. (1)(2) and (10), R_{LRS} is designed with 99 Ω for k = 1 and 1.0×10^5 Ω for k = 1024 when r_{BL} = 1 Ω to obtain $I_{\mathrm{BL\_CiM\_signal}}$ with α = 0.99 of the ideal r_{BL} = 0 Ω case. Similarly, R_{LRS} is designed for different r_{BL} values.

Step 2: $I_{\mathrm{BL\_CiM\_offset}}$ design (Fig. 10) To eliminate the $I_{\mathrm{BL\_CiM\_offset}}$ from the $I_{\mathrm{BL\_CiM\_signal}}$, β in Eq. (11) is determined. In this work, β is set to 0.1, 0.2, and 0.5. The lower β distinguishes the signal current $I_{\mathrm{BL\_CiM\_signal}}$ from the offset current $I_{\mathrm{BL\_CiM\_offset}}$. If β = 1.0, $I_{\mathrm{BL\_CiM\_offset}}$ is equal to $I_{\mathrm{BL\_CiM\_signal}}$. For the best ($k$ = 1, Fig. 10(a)) and worst (k = 1024, Fig. 10(b)) cases, $I_{\mathrm{BL\_CiM\_offset}}$ is determined for each r_{BL} value. A higher r_{BL} requires a smaller $I_{\mathrm{BL\_CiM\_offset}}$. Thus, as determined in Step 3, a higher R_{HRS} is required to reduce $I_{\mathrm{BL\_CiM\_offset}}$.

Step 3: R_{HRS} design (Fig. 11) The ReRAM HRS resistance R_{HRS} satisfying Eq. (11) is designed for the best (k = 1, Fig. 11(a)) and worst (k = 1024, Fig. 11(b)) cases of 1 LRS cell position k, the BL resistance r_{BL}, and the indicator β of $I_{\mathrm{BL\_CiM\_offset}}$ elimination. Fig. 11 shows R_{LRS} and R_{HRS} for each r_{BL} and β. R_{LRS} in Fig. 11 are determined at Step 1. A higher R_{HRS} is required to reduce $I_{\mathrm{BL\_CiM\_offset}}$ by N HRS cells. If the BL resistance r_{BL} is reduced from 1 Ω to 0.25 Ω by multi-level interconnection, R_{HRS} value can be reduced to about 1/10.

5. Conclusion

This paper gives mathematical formulations of the BL current affected by the BL resistance r_{BL} for the digital ReRAM memory and the analog ReRAM CiM. Fig. 12 summarizes the ReRAM resistance design of LRS and HRS with two capacity scenarios. The low-capacity CiM divides the memory cell array into smaller arrays, but the area of the extra peripheral circuits is large (Fig. 12(a)). The high-capacity CiM has a large memory cell array with small area of peripheral circuits (Fig. 12(b)). The high-capacity CiM has larger number of memory cells N connected to one BL. Fig. 12(c) summarizes the ReRAM resistance design. The ReRAM LRS resistance R_{LRS} of the digital ReRAM memory should be high to achieve high capacity when reading k =



(c) r_{BL} = 1 Ω, α = 0.99, β = 0.1

		Low capacity	High capacity
Digital ReRAM Memory	Memory number k to be read	16	1024
	ReRAM LRS resistance R_{LRS} [Ω]	1.6×10^3	1.0×10^5
Analog ReRAM CiM	Number of connected memories N	16	1024
	Memory resistance of R_{LRS} and R_{HRS} [Ω] in best and worst cases	99 and 1.6×10^4 1.6×10^3 and 2.6×10^5	99 and 7.0×10^5 1.0×10^5 and 1.0×10^9

Fig. 12 Capacity scenarios of analog ReRAM CiM for (a) low capacity and (b) high capacity. (c) ReRAM resistance design of LRS and HRS for digital ReRAM memory and analog ReRAM CiM.

1024th cell in N = 1024 memory cells. For the analog ReRAM CiM, the ReRAM resistance of LRS and HRS (R_{LRS} and R_{HRS}) is set to distinguish the cell current of 1 LRS cell from that of N HRS cells. As a worst case, 1 LRS cell is stored at the farthest ReRAM cell from the ADC in a BL. 1 LRS cell is stored in k = 16th (1024th) cell for N = 16 (1024) of the low- (high-) capacity CiM scenario. R_{LRS} and R_{HRS} of the analog ReRAM CiM are 1.6×10^3 Ω and 2.6×10^5 Ω for the low-capacity analog ReRAM CiM (N = 16), and 1.0×10^5 Ω and 1.0×10^9 Ω for the high-capacity analog ReRAM CiM (N = 1024). The recent studies show a resistance range from 10^3 Ω for LRS to 10^8-10^9 Ω for HRS in TaO_X-based ReRAM cells [27, 28]. Thus, the ReRAM resistance design in Fig. 12(c) is feasible for both digital ReRAM memory and analog ReRAM CiM.

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Appendix

The derivation of Eqs. (7) and (8) from Eqs. (4)-(6) is shown. A matrix representation of the cell current i_{k+1} and BL current j_{k+1} is obtained by eliminating the local BL voltage v_k and v_{k+1} from Eqs. (4)-(6).

$$\begin{pmatrix} 0 & 1 \\ R & r_{\mathrm{BL}} \end{pmatrix} \begin{pmatrix} i_{k+1} \\ j_{k+1} \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ R & 0 \end{pmatrix} \begin{pmatrix} i_k \\ j_k \end{pmatrix} \quad \text{(A1)}$$

$$\begin{pmatrix} i_{k+1} \\ j_{k+1} \end{pmatrix} = \begin{pmatrix} {r_{\mathrm{BL}}}/{R} + 1 & -{r_{\mathrm{BL}}}/{R} \\ -1 & 1 \end{pmatrix} \begin{pmatrix} i_k \\ j_k \end{pmatrix} \tag{A2}$$

Here, let $\gamma = {r_{\mathrm{BL}}}/{R}$ and define a matrix A.

$$\begin{pmatrix} i_{k+1} \\ j_{k+1} \end{pmatrix} = \begin{pmatrix} \gamma + 1 & -\gamma \\ -1 & 1 \end{pmatrix} \begin{pmatrix} i_k \\ j_k \end{pmatrix} \equiv A \begin{pmatrix} i_k \\ j_k \end{pmatrix} \tag{A3}$$

$$A \equiv \begin{pmatrix} \gamma + 1 & -\gamma \\ -1 & 1 \end{pmatrix} \tag{A4}$$

Then, using the first cell current i_1 and BL current j_1 at the end of BL, i_k and j_k related to k-th memory cell are represented as Eq. (A5).

$$\begin{pmatrix} i_k \\ j_k \end{pmatrix} = A \begin{pmatrix} i_{k-1} \\ j_{k-1} \end{pmatrix} = \cdots = A^{k-1} \begin{pmatrix} i_1 \\ j_1 \end{pmatrix} \tag{A5}$$

Where $\delta = \sqrt{\gamma^2 + 4\gamma}$, λ_1 and λ_2 indicate the eigenvalues of matrix A. The corresponding eigenvectors $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ of matrix A are also obtained.

$$\lambda_1 = \frac{(\gamma - \delta + 2)}{2}, \quad \lambda_2 = \frac{(\gamma + \delta + 2)}{2} \tag{A6}$$

$$\boldsymbol{x}_1 = \begin{pmatrix} {(-\gamma + \delta)}/{2} \\ 1 \end{pmatrix}, \quad \boldsymbol{x}_2 = \begin{pmatrix} {(-\gamma - \delta)}/{2} \\ 1 \end{pmatrix} \tag{A7}$$

The matrix A is represented by a diagonal matrix as follows.

$$\begin{aligned} A &= SJS^{-1} \\ &= \begin{pmatrix} -\lambda_1 + 1 & -\lambda_2 + 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \frac{1}{\delta} \begin{pmatrix} 1 & \lambda_2 - 1 \\ -1 & -\lambda_1 + 1 \end{pmatrix} \end{aligned} \tag{A8}$$

Then, Eq. (A5) is simplified as follows.

$$\begin{pmatrix} i_k \\ j_k \end{pmatrix} = A^{k-1} \begin{pmatrix} i_1 \\ j_1 \end{pmatrix} = SJ^{k-1}S^{-1} \begin{pmatrix} i_1 \\ j_1 \end{pmatrix} \tag{A9}$$

The conditions for solving Eq. (A9) is obtained as Eq. (A10) since $v_1 = 0$ (grounded) and $j_{N+1} = 0$ (no current).

$$i_1 = \frac{V_{\mathrm{SL}}}{R} \text{ and } j_{N+1} = 0 \tag{A10}$$

By applying the condition in Eq. (A10), Eq. (A9) becomes

$$\begin{pmatrix} i_{N+1} \\ 0 \end{pmatrix} = SJ^N S^{-1} \begin{pmatrix} {V_{\mathrm{SL}}}/{R} \\ j_1 \end{pmatrix}. \tag{A11}$$

Here, each element of the matrix SJ^NS^{-1} is represented as Eq. (A12).

$$\begin{aligned} {SJ^NS^{-1}}_{11} &= \frac{1}{\delta}\{-(\lambda_1 - 1)\lambda_1^N + (\lambda_2 - 1)\lambda_2^N\} \\ {SJ^NS^{-1}}_{12} &= -\frac{1}{\delta}\{(\lambda_1 - 1)(\lambda_2 - 1)(\lambda_1^N - \lambda_2^N)\} \\ {SJ^NS^{-1}}_{21} &= \frac{1}{\delta}(\lambda_1^N - \lambda_2^N) \\ {SJ^NS^{-1}}_{22} &= \frac{1}{\delta}\{(\lambda_2 - 1)\lambda_1^N - (\lambda_1 - 1)\lambda_2^N\} \end{aligned} \tag{A12}$$

From the second element of the vector of Eq. (A11), j_1 as the BL current $I_{\mathrm{BL\_CiM}}$ of the analog ReRAM CiM is obtained in Eq. (A13) with the parameters in Eq. (A14) same as Eqs. (7) and (8).

$$\begin{aligned} I_{\mathrm{BL\_CiM}} &= j_1 \\ &= -\frac{\lambda_1^N - \lambda_2^N}{(\lambda_2 - 1)\lambda_1^N - (\lambda_1 - 1)\lambda_2^N} \frac{V_{\mathrm{SL}}}{R} \end{aligned} \tag{A13}$$

$$\begin{gathered} \lambda_1 = \frac{(\gamma - \delta + 2)}{2}, \quad \lambda_2 = \frac{(\gamma + \delta + 2)}{2}, \\ \delta = \sqrt{\gamma^2 + 4\gamma}, \quad \gamma = \frac{r_{\mathrm{BL}}}{R} \end{gathered} \tag{A14}$$

Fig. A1 shows the eigenvalues λ_1, λ_2 and eigenvectors $\boldsymbol{x}_1$, $\boldsymbol{x}_2$ calculated with Eqs. (A6) and (A7) with respect to γ. The

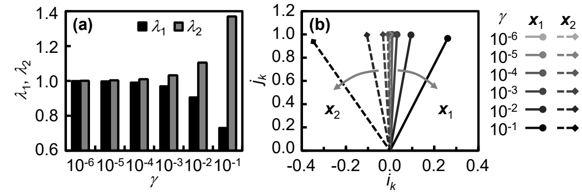

Fig. A1 (a) Eigenvalues λ_1, λ_2 and (b) eigenvectors $\boldsymbol{x}_1$, $\boldsymbol{x}_2$ of matrix A with respect to $\gamma = r_{\mathrm{BL}}/R$.

eigenvalues λ_1, λ_2 are almost 1 when γ is small. The larger γ enlarges the difference between λ_1 and λ_2 (Fig. A1(a)). When γ is small, the eigenvectors $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ lie almost in (0, 1). However, with the larger γ, $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ corresponding to λ_1 and λ_2 rotate clockwise and counterclockwise, respectively (Fig. A1(b)). The eigenvector $\boldsymbol{x}_2$ is dominant since the eigenvalue λ_2 is larger than λ_1. Thus, the initial value (i_1, j_1) gradually approaches to the direction of $\boldsymbol{x}_2$ as they are iteratively transformed by the matrix A in Eq. (A5). When the large number N of cells connected to BL, (i_k, j_k) converges in the $\boldsymbol{x}_2$ direction. In this case, the negative i_k value implies that the cell current flows backward due to the BL resistance r_{BL}.

Chihiro Matsui is currently a Project Associate Professor in the Department of Electronics Engineering and Information Systems, Graduate School of Engineering of The University of Tokyo. Her research interest includes system, circuit, and device co-design with emerging non-volatile memories for enterprise applications. She earned her B.S. and M.S. degrees in Physics from Ochanomizu University, Tokyo, Japan, in 2003 and 2005, respectively, and her Ph.D. degree in Information Security Sciences from Chuo University, Tokyo, Japan, in 2018. She was a Project Assistant Professor of Research and Development Initiative at Chuo University from 2018 to 2020 and a Project Assistant Professor in the Department of Electronics Engineering and Information Systems, Graduate School of Engineering of The University of Tokyo from 2020 to 2023.

Ayumu Yamada received the B.E. degree in Electrical Engineering and the M.E. degree in Electrical Engineering and Information Systems from the University of Tokyo in 2022 and 2024, respectively. His current research interests include Computation-in-Memory (CiM), emerging non-volatile memories, neuromorphic computing, and Bayesian machine learning.

Naoko Misawa received the M.S. degree from Imperial College London in 2012. She is currently an academic staff in Takeuchi Laboratory in the department of Electrical Engineering and Information Systems, Graduate School of The University of Tokyo. Her research interests include emerging non-volatile memories, neuromorphic computing, and Vision Transformer.

Ken Takeuchi is a Professor at Department of Electrical Engineering and Information Systems of The University of Tokyo. He is now working on data-centric computing such as computation in memory, approximate computing, AI chip design and brain-inspired memory. He received B.S. and M.S. degrees in Applied Physics and Ph.D. degree in Electric Engineering from The University of Tokyo in 1991, 1993 and 2006, respectively. In 2003, he received M.B.A. degree from Stanford University. Since he joined Toshiba in 1993, he led Toshiba's NAND flash memory circuit design for fourteen years. He was an Associate Professor at Department of Electrical Engineering and Information Systems of The University of Tokyo from 2007 till 2012. He was a Professor at Department of Electrical, Electronic and Communication Engineering of Chuo University from 2012 till 2020. In 2020, he rejoined The University of Tokyo. At Toshiba, he designed six world's highest density NAND flash memory products such as 0.7um 16Mbit, 0.4um 64Mbit, 0.25um 256Mbit, 0.16um 1Gbit, 0.13um 2Gbit and 56nm 8Gbit NAND flash memories. He holds 229 patents worldwide. With his invention, “multipage cell architecture”, presented at Symposium on VLSI Circuits in 1997, he commercialized world's first multi-level cell NAND flash memory in 2001. He authored numerous technical papers which won the Takuo Sugano Award for Outstanding Paper at ISSCC 2007, the Silkroad Award at ISSCC 2015 and Best Demo Paper Award at Symposium on VLSI Circuits in 2021. Since 2018 he has been serving as an executive committee member of Symposium on VLSI Technology & Circuits. He served as the symposium chair/co-chair of Symposium on VLSI Circuits in 2021/2020. He served as the program chair/co-chair of Symposium on VLSI Circuits in 2019/2018. He served as the program chair/co-chair of Asian Solid-State Circuits Conference (A-SSCC) in 2023/2022. He has served on the program committee member of International Solid-State Circuits Conference (ISSCC), Custom Integrated Circuits Conference (CICC), Asian Solid-State Circuits Conference (A-SSCC), International Memory Workshop (IMW) and International Conference on Solid State Devices and Materials (SSDM). He served as the guest editor for IEEE Journal of Solid-State Circuits and associate editor of IEEE Solid-State Circuits Letters.